In our 3-tier architecture, private EC2 instances were accessed using AWS Systems Manager Session Manager. To enable this, the instances must be able to communicate with the AWS Systems Manager (SSM) service.

Our Architecture

We have:

- EC2 instances in private subnets
- 2 NAT Gateways
- Access via Session Manager

That means our private instances already have outbound access like this:

```
EC2 (private subnet)
      ↓
Private Route Table
      ↓
NAT Gateway
      ↓
Internet Gateway
      ↓
AWS Systems Manager service
```

So, Session Manager will already work through the NAT Gateway.

Therefore:

We do NOT strictly need VPC endpoints for SSM if NAT Gateway exists.

Requirements for SSM to Work

Our EC2 instance has:

IAM Role

Attach the role:

AmazonSSMManagedInstanceCore

In our architecture

Because we already have 2 NAT Gateways, our private instances can already reach:

- SSM
- S3
- patch repositories
- package managers

So, Session Manager will work without endpoints.

For our architecture:

- We do NOT strictly need EC2-to-VPC endpoints for Session Manager, because our private instances can reach SSM through the NAT Gateways.
- However, creating VPC endpoints for SSM is considered a best-practice improvement because it keeps management traffic entirely inside AWS.

Our EC2 has an IAM role attached

Our instances have an **instance profile** with at least **AmazonSSMManagedInstanceCore** so they can talk to Systems Manager.

In the AWS console:

- EC2 → Instances → select the instance
- Actions → Security → **Modify IAM role**
- Attach the role that contains **AmazonSSMManagedInstanceCore**

Because our instances live in a **private subnet**, they must have outbound connectivity either through:

- a **NAT Gateway**, or
- **VPC endpoints** for `ssm`, `ssmmessages`, and `ec2messages`.

For our setup:

- the private subnet route table has `0.0.0.0/0` to the correct **NAT Gateway**, or
- the SSM VPC endpoints exist in the VPC and their security groups allow traffic from the instance

How Session Manager connects

Traffic path:

EC2 Instance
↓
SSM Agent
↓
HTTPS port 443
↓
AWS Systems Manager service
↓
Session Manager terminal

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough | Do not show me this message again.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

citymart-backend-server-ami

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-087c9ba923d9765d8 (64-bit (x86), uefi-preferred) / ami-0f51f93203f42fefe (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260302.1 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (...	uefi-preferred	ami-087c9ba923d9765d8	2026-03-03	ec2-user

Verified provider

Instance type Info | Get advice

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0118 USD per Hour
On-Demand SUSE base pricing: 0.0118 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0153 USD per Hour
On-Demand RHEL base pricing: 0.0406 USD per Hour
On-Demand Windows base pricing: 0.021 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

citymart-backend-server-keypair

Create new key pair

Summary

Number of instances Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-087c9ba923d9765d8

Virtual server type (instance type)
t3.micro

Cancel | Launch instance

Preview code

Network settings Info

VPC - *required* | Info

vpc-0206bba22a724b50b (citymart-3-tier-architecture-vpc)
10.0.0.0/16

Subnet | Info

subnet-0c8a3c4c0ae7b2d7a app-private-subnet-3a
VPC: vpc-0206bba22a724b50b Owner: 009850210027
Availability Zone: eu-west-2a (euw2-az2) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.96.0/20)

Create new subnet

Auto-assign public IP | Info

Disable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- ○ Create security group
- ◉ Select existing security group

Common security groups | Info

Select security groups

Compare security group rules

citymart-backend-server-sg sg-0194fdd9bc542789f ✕
VPC: vpc-0206bba22a724b50b

Security groups that you add or remove here will be added to or removed from all your network interfaces.

▶ **Advanced network configuration**

Configure storage Info

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems Edit

Advanced details Info

Domain join directory | Info

Select

Create new directory

IAM instance profile | Info

3-tier-architecture-sam
arn:aws:iam::009850210027:instance-profile/3-tier-architecture-sam

Create new IAM profile

Hostname type | Info

IP name

DNS Hostname | Info

- [x] Enable IP name IPv4 (A record) DNS requests
- [ ] Enable resource-based IPv4 (A record) DNS requests
- [ ] Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery | Info

Select

Shutdown behavior | Info

Stop

Stop - Hibernate behavior | Info

Select

Termination protection | Info

Select

Stop protection | Info

Select

Detailed CloudWatch monitoring | Info

Select

Credit specification | Info

Unlimited

Additional charges apply

Placement group | Info

Select

Create new placement group

EBS-optimized instance | Info

Enable

Instance bandwidth configuration | Info

Select

Purchasing option | Info

- (●) None
- ( ) Capacity Blocks
  Launch instances for your active capacity blocks
- ( ) Interruptible Capacity Reservations
  Launch instances for your interruptible Capacity Reservations
- ( ) Spot instances
  Request Spot Instances at the Spot price, capped at the On-Demand price

Capacity reservation | Info

Select

Tenancy | Info

Select

RAM disk ID | Info

Select

Kernel ID | Info

Select

Nitro Enclave | Info

Select

License configurations | Info

Select

CPU options - *update* | Info

Configure CPUs for your instance to optimize performance and save on licensing costs.

- (●) Use default CPU options
- ( ) Specify CPU options

Default active vCPUs	Total vCPUs
2	2

Nested virtualization | Info

Select

Nested virtualization is not supported with the selected instance type.

Metadata accessible | Info

Enabled

Metadata IPv6 endpoint | Info

Select

Metadata version | Info

V2 only (token required)

⚠ For V2 requests, you must include a session token in all instance

metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit Info

2

Allow tags in metadata Info

Select

User data - *optional* Info

Upload a file with your user data or enter it in the field.

Choose file

User data has already been base64 encoded

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Amazon Linux 2023 AMI 2023.10....read more
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Cancel | Launch instance

Preview code

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VPC - *required* | Info

vpc-0206bba22a724b50b (citymart-3-tier-architecture-vpc)
10.0.0.0/16

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EC2 > Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances (2) Info

Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states

1

Name	Instance ID	Instance state	Instance type	Status check	Alarm stat
citymart-frontend-server-ami	i-08dd11760e00e31c0	Running	t3.micro	3/3 checks passed	View alarm
citymart-backend-server-ami	i-0887d9ee89f92b8e7	Running	t3.micro	3/3 checks passed	View alarm

aws | Search [Alt+S] | Europe (London) | Samuel A. O. (0098-5021-0027) cba_admin

EC2 > Instances > i-0887d9ee89f92b8e7 > Connect to instance

Instance ID	VPC ID	Security groups	IAM role
i-0887d9ee89f92b8e7 (citymart-backend-server-ami)	vpc-0206bba22a724b50b	sg-0194fdd9bc542789f (citymart-backend-server-sg)	3-tier-architecture-sam

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

SSM agent status

Introducing Systems Manager just-in-time node access
Move towards zero standing privileges by requiring operators to request access before remotely connecting to instances. Learn more

Try for free

Ping status	Last ping time	Session Manager connection status	Agent version
Online	Sun Mar 08 2026 13:04:19 GMT+0200 (Eastern European Standard Time)	Connected	3.3.3598.0

CloudShell | Feedback | Console Mobile App

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